b) B_1 is pushed towards B_2.
c) I_1 is decreased.
d) I_1 is increased.

8 Use Lenz's Law to specify the direction of the induced current in the loop and in the moving conductor in each of the cases shown in figure 8.22.

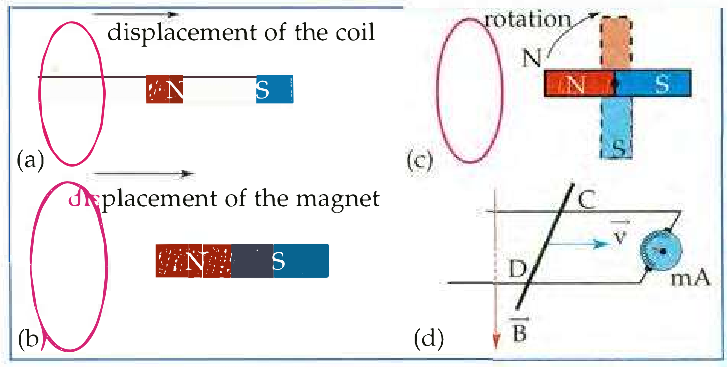

Figure 8.22

PROBLEMS

9 A loop of surface area s_1 = 10 cm², placed in a uniform magnetic field $\vec{B}_1$ of magnitude $B_1 = 8 \times 10^{-2}$T, has its plane perpendicular to the field lines. A second loop of area s_2 = 15 cm² is placed in another uniform field $\vec{B}_2$ of magnitude B_2 = 0.1T such that its normal unit vector $\vec{n}$ makes an angle $\theta = 30°$ with the lines of field $\vec{B}_2$.

1) Which loop receives more magnetic flux?
2) Are there any induced currents in these two loops? Justify.

10 A square loop of side a = 10 cm and resistance R=0.1Ω is placed in a uniform magnetic field $\vec{B}$ whose magnitude varies with time as shown in figure 8.23.

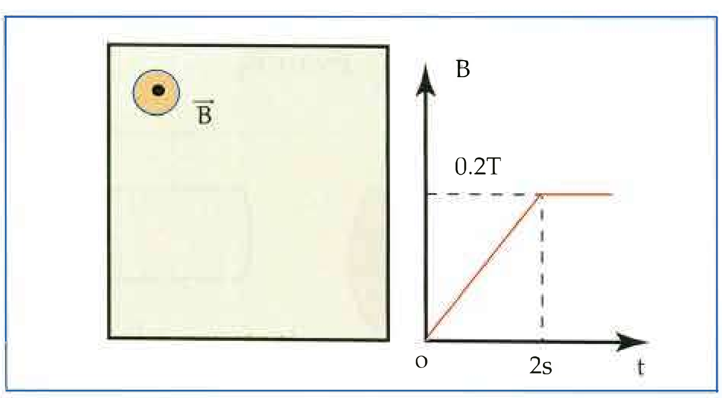

Figure 8.23

a) Determine, by applying Lenz's Law, the direction of the induced current for 0 < t < 2s, and for t > 2 sec.
b) After choosing a positive direction along the loop, write down the expression of B as a function of time and calculate, the values of the magnetic flux crossing the loop for 0 < t < 2s and for t > 2s.
c) Determine the induced e. m. f, the direction and the magnitude of the induced current. Compare the direction of the induced current to that found in part "a" above.
d) Verify that the direction of the induced current does not depend on the arbitrary choice of the positive direction.

11 A rectangular loop of area S is placed in uniform magnetic field $\vec{B}$ whose magnitude B varies with time .We choose the positive direction as in figure 8.24. The two terminals A and C of the loop are connected to one channel of an oscilloscope.

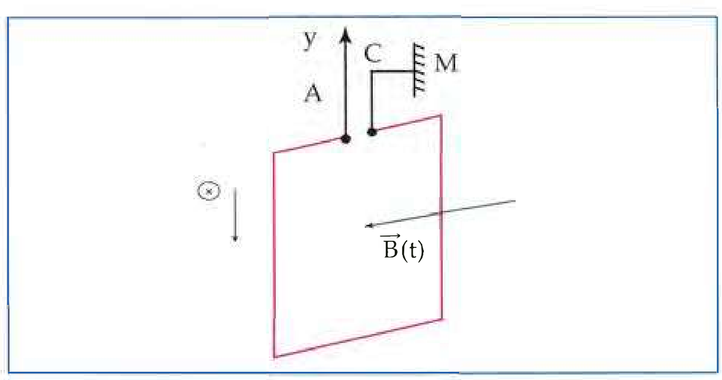

Figure 8.24